



DEPI-SALES PERFORMANCE - METRICS !

Understanding Sales Performance Metrics & KPIs

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A. Market size & Growth Trends

Introduction:

This section provides an analytical overview of the retail industry based on the dataset Different_stores_data_V2. The objective is to evaluate the overall market size, examine historical growth trends, identify seasonal or cyclical sales patterns, and highlight regional variations. By understanding these dimensions, decision-makers can assess how the market has evolved over time and which periods or regions present the greatest business potential.

1. Total Market Size (Sales Volume or Revenue)

The total market size represents the overall value of sales generated within a specific period. It is calculated by summing the revenue of each transaction using the formula:

$$\text{Revenue} = \text{Selling Price} \times \text{Quantity Sold.}$$

This indicator provides a clear picture of how large the market currently is in terms of sales value. When analyzing the dataset, the total revenue can be aggregated either monthly or annually to show how the market performs across different time frames.

2. Historical Growth (Past 3–5 Years)

To understand market development over time, historical growth is analyzed by comparing annual revenue values over the past three to five years. The year-over-year (YoY) growth rate can be calculated as:

$$\text{Growth Rate} = ((\text{Revenue in Current Year} - \text{Revenue in Previous Year}) / \text{Revenue in Previous Year}) \times 100.$$



This metric reveals whether the market is expanding, stable, or contracting. A consistent upward trend indicates a healthy, growing market, while stagnation or decline may suggest the need for innovation or market diversification.

3. Seasonality or Cyclicity of Sales

Seasonality analysis identifies recurring sales patterns throughout the year. By extracting the month from the `invoice_date` column and calculating the total monthly revenue, the analysis highlights which months experience the highest or lowest sales volumes.

If certain months consistently show revenue spikes, these can be considered peak seasons — ideal for targeted marketing campaigns, discount strategies, or stock increases. For example, higher sales may coincide with holidays, back-to-school periods, or national shopping events. Conversely, months with weaker performance might indicate a need for promotional efforts or product repositioning.

4. Regional or Country-Specific Insights

Regional analysis provides insights into how different locations contribute to total sales. Using the variables `region`, `state`, and `shopping_mall`, revenue can be grouped and compared geographically. The percentage contribution of each region to total revenue reveals which markets are the strongest and where growth potential exists.

Some regions might show a smaller number of transactions but higher total revenue, indicating that customers in those areas purchase higher-value products. Visualizing regional sales performance through bar charts or heat maps can make these insights more intuitive. Such analysis helps in prioritizing marketing resources and optimizing inventory distribution according to regional demand.



Conclusion and Key Takeaways

The “Market Size and Growth Trends” analysis serves as a foundation for understanding the market’s current position and future trajectory. Based on this analysis, several strategic recommendations can be drawn:

- ❖ Focus marketing efforts on high-performing months to capitalize on seasonal peaks.
- ❖ Reallocate inventory and marketing resources toward regions with higher profitability or growth potential.
- ❖ Continuously monitor annual growth rates and extend the analysis over five years to enhance forecasting accuracy and support long-term planning.

KPI	Definition	Formula
Top-Selling Category	the product category that achieved the highest total sales volume	$\sum(\text{Selling Price} \times \text{Quantity Sold})$
Category Profitability	It is measuring the total profit generated by each product category.	$\sum[(\text{Selling Price} - \text{Cost Price}) \times \text{Quantity Sold}]$
Average Quantity Sold per Category	It is calculating the average number of units sold for each product category.	$\sum(\text{Quantity Sold}) / \text{Number of products in category}$
Category Revenue Contribution	It is showing each category’s percentage contribution to the total company revenue	$(\text{Revenue of Category} / \text{Total Revenue}) * 100$
Average Markup per Category	It is indicating how much higher the selling price is compared to the cost price, expressed as a percentage	$[(\text{Selling Price} - \text{Cost Price}) / \text{cost price}] * 100$



Competitive Landscape in E-commerce

Scope: Global overview with emphasis on MENA/Egypt regional players and recent competitive dynamics.

Executive summary

E-commerce remains the dominant growth channel for retail worldwide. Market forecasts show continued high growth into the 2030s, while a small set of global platforms (Amazon, Alibaba) hold outsized shares in their home markets. New low-cost challengers (e.g., Temu) and strong regional players (Noon, Jumia, Carrefour online, Talabat/Delivery platforms) are reshaping pricing, logistics, and payment expectations — especially in MENA markets where digital payments and mobile adoption are accelerating.

1. Market overview (key numbers & trends)

- ✦ **Global scale & growth:** Market analyses project multi-trillion USD totals with double-digit CAGRs through 2030 (estimates vary by source; one widely cited forecast shows a very large market expanding through the decade). Growth drivers are mobile commerce, embedded finance (BNPL/digital wallets), and AI personalization.
- ✦ **MENA & Egypt:** MENA e-commerce is a fast-growing region with strong investment in payments and logistics; estimates for the Middle East & Africa point to a multi-hundred-billion opportunity over the next 5–8 years. Egypt's GMV is forecast to reach multiple billions USD (Egypt expected to be a high-growth national market in the region).



2. Who are the leaders (profiles & strengths)

Global leaders

- ✦ **Amazon** — dominant in the U.S. with ~37_38% of U.S. e-commerce spend; strength from Prime ecosystem, marketplace scale, logistics network, and advertising. Its marketplace effects set baseline customer expectations on price, delivery speed and returns.
- ✦ **Alibaba / Taobao / Tmall** — dominance in mainland China via integrated commerce + payments + logistics + social commerce; massive GMV and ecosystem play.
- ✦ **Walmart** — strongest omnichannel challenger in the U.S., leveraging physical stores + pickup/fulfillment.

Global challengers / disruptors

- ✦ **Temu (PDD/Shein-style entrants)** — aggressive low-price, high-marketing model targeting developed markets; heavy ad spend and promotional pricing; faces sustainability and margin questions but is an immediate competitive pressure on price. Recent reporting shows Temu investing heavily in ads but struggling to match Amazon's pricing power in some markets.

Regional leaders (MENA / Egypt examples)

- ✦ **Noon (UAE / KSA / Egypt presence)** — local UX, marketplace + first-party, fast delivery in core GCC markets.



- † **Jumia (Africa, including North Africa)** — regional network focused on last-mile challenges and cash-on-delivery options.
- † **Carrefour (online) / Majid Al Futtaim** — omnichannel grocery and general merchandise; strong local logistics/inventory integration.
- † **Talabat & regional delivery apps** — control local quick-commerce and food delivery; increasingly used for grocery & convenience commerce. (Regional market commentary and growth figures specific to Egypt and MENA are reflected in national market reports.)

3. Competitive dynamics — what actually differentiates winners

- † **Ecosystem & stickiness** — marketplaces that bundle subscriptions, payments, ads and logistics (Amazon, Alibaba) generate higher customer lifetime value and barrier to exit.
- † **Logistics & fulfillment** — speed, reliability, returns handling and fulfillment density are primary battlegrounds; local couriers and dark-store networks matter in MENA.
- † **Price & assortment** — low-cost imports (Temu, Shein) pressure margins; brand/value differentiation, localized assortments and authenticity claims counter pure price plays.
- † **Payments & trust** — cash-on-delivery remains important in parts of MENA & Africa; accelerating card/Wallet/Buy-Now-Pay-Later penetration improves conversion.
- † **Regulation & trade rules** — import duties, de-minimis changes, and local rules can rapidly change competitive economics (recent policy shifts have affected cross-border low-value imports).



4. Opportunities & risks for a business entering or scaling in e-commerce

Opportunities

- ✦ **Regional expansion via marketplaces:** Fast reach with lower setup cost; combine marketplace presence with owned DTC channel to capture margin uplift).
- ✦ **Mobile & quick commerce:** Leverage mobile UX and instant-commerce (dark stores/delivery partners) to win frequency.
- ✦ **Embedded finance & loyalty:** Integrate BNPL/digital wallets and subscription models to increase AOV and retention.

Risks

- ✦ **Price wars and low-margin challengers:** Heavy discounting from low-cost entrants can erode margins.
- ✦ **Logistics & returns cost:** High return rates or poor last-mile raise unit economics risk in low-AOV categories.
- ✦ **Regulatory shocks:** Sudden customs/tax changes affecting cross-border commerce may disrupt business models.

5. Strategic recommendations (actionable)



1. **Hybrid channel strategy:** List on 1-2 dominant marketplaces locally (Noon/Jumia/Amazon where relevant) while investing in a conversion-optimized DTC site to capture data & margin.
2. **Prioritize fulfilment reliability over lowest price:** Use regional 3PL partnerships or micro-fulfilment to reduce lead times and returns; advertise “next-day” where viable.
3. **Segment portfolio by margin & channel fit:** Push high-margin/brand SKUs via DTC and marketplaces for volume SKUs; use selective promotions to protect margin.
4. **Invest in payments & local trust signals:** Offer preferred local payment methods (COD where needed), clear returns and authenticity messaging.
5. **Monitor emerging low-cost entrants and regulatory changes:** Keep a dedicated competitive watch on players like Temu and on customs/tariff policy updates affecting cross-border economics.

6. Suggested KPIs to track (dashboard-ready)

- ✦ Gross Merchandise Value (GMV) by channel & country.
- ✦ Share of GMV by marketplace vs DTC.
- ✦ Customer Acquisition Cost (CAC) & Lifetime Value (LTV) per channel.
- ✦ Fulfillment cost per order & average delivery time.
- ✦ Return rate (%) and return cost per order.
- ✦ Average Order Value (AOV) and conversion rate by channel



Sales Channels & Distribution

Common Channels in Retail & E-commerce:

- 1) **Online Marketplaces:** Platforms like Amazon, Jumia, and Noon dominate e-commerce in MENA, offering broad reach and low entry barriers.
- 2) **Retail Stores:** Brick-and-mortar outlets remain vital for experiential shopping, especially in malls and high-traffic urban zones.
- 3) **Wholesalers:** Serve as bulk suppliers to smaller retailers, often regionally segmented.
- 4) **Distributors:** Act as intermediaries between manufacturers and retailers, crucial for scaling across multiple regions.

Regional Sales Metrics

KPI	Definition	SQL/DAX Logic
Sales by Region	Total revenue per region	<code>SUM (quantity * selling_price_per_unit) GROUP BY region</code>
Profit by Shopping Mall	Revenue minus cost per mall	<code>SUM ((selling_price_per_unit - cost_price_per_unit) * quantity) GROUP BY shopping_mall</code>
Top Performing State	State with highest total profit	Rank states by total profit using same formula as above
Average Transaction Value per Region	Avg. revenue per invoice	<code>SUM(selling_price_per_unit * quantity) / COUNT(DISTINCT invoice_no) GROUP BY region</code>
Regional Profit Margin	Profit as % of revenue	<code>SUM(profit) / SUM(revenue) GROUP BY region</code>



Consumer Behavior & Trends

Introduction

Consumer behavior plays a central role in determining the overall success of a business's sales performance. It involves understanding how individuals make decisions to spend their available resources—time, money, and effort—on purchasing goods and services. In a competitive market, analyzing consumer preferences and behavioral trends helps organizations align their marketing and sales strategies to meet customer expectations, optimize profitability, and enhance customer retention.

1. Customer Preferences

- ✦ Customer preferences reflect the specific choices consumers make when selecting products, brands, or services. These preferences are shaped by factors such as product quality, price, brand image, and convenience.
- ✦ In the sales industry, customer preferences are often dynamic. For instance, some consumers prioritize affordability, while others focus on brand reputation or sustainability. Businesses must continually monitor and analyze purchasing patterns—such as the most frequently purchased categories, preferred payment methods, and average spending per visit—to identify shifts in consumer taste.

Understanding customer preferences allows companies to:

- ✦ Tailor marketing campaigns to match customer needs.
- ✦ Optimize product assortments based on demand.
- ✦ Enhance customer experience by offering personalized recommendations.

Ultimately, aligning sales strategies with consumer preferences leads to improved conversion rates, higher customer satisfaction, and stronger sales performance.



2. Factors Influencing Buying Decisions

The buying process is affected by a combination of **psychological**, **personal**, **social**, and **situational** factors:

- ✦ **Psychological Factors:** Motivation, perception, and attitudes shape how customers evaluate products. For example, consumers motivated by self-expression might prefer premium or branded products, while others motivated by practicality may prioritize discounts and functionality.
- ✦ **Personal Factors:** Age, gender, income, and lifestyle influence what and how often consumers buy. Younger audiences may favor trendy or technology-driven products, whereas older consumers may focus on reliability and value.
- ✦ **Social Factors:** Family, friends, and social media networks play a powerful role in shaping opinions. Recommendations and online reviews can directly impact sales conversions.
- ✦ **Situational Factors:** Contextual elements such as seasonality, time constraints, promotions, or location also affect purchase behavior. For instance, sales tend to rise during festive seasons or promotional events.

Businesses that understand and act on these influencing factors can design more effective marketing strategies, improve product positioning, and ultimately enhance their overall sales performance.



3. Emerging Consumer Trends

Consumer trends evolve continuously as technology, society, and global conditions change. Several key trends currently define consumer behavior in the sales industry:

1. **Personalization and Data-Driven Experiences** – Modern consumers expect tailored offers and product recommendations. Companies are leveraging data analytics to understand individual preferences and provide customized shopping experiences.
2. **Omnichannel Shopping Behavior** – Shoppers now move fluidly between online and physical stores. Businesses that integrate their sales channels effectively can boost convenience and overall revenue.
3. **Value-Conscious Purchasing** – Economic uncertainty has increased consumers' sensitivity to price and value. Promotions, loyalty programs, and flexible payment options influence purchase decisions more than ever.
4. **Sustainability and Ethical Consumption** – Environmentally aware consumers prefer eco-friendly and socially responsible brands. Businesses emphasizing ethical sourcing and sustainability are gaining competitive advantages.
5. **Rise of Digital and Mobile Payments** – Mobile wallets and digital payment systems have made transactions faster and more convenient, influencing where and how consumers choose to buy.

Recognizing and adapting to these emerging trends enables businesses to maintain strong sales performance in rapidly changing markets.



4. Demographics and Target Audience Insights

Demographic insights provide a foundation for understanding who the customers are and how their characteristics affect buying behavior. Key demographic variables include **age, gender, income level, occupation, and region**.

For example:

- **Age:** Younger customers may purchase more frequently but with lower average order values, while older customers may spend more per transaction.
- **Gender:** Certain product categories show clear gender-based preferences, influencing marketing and product strategies.
- **Region and State:** Geographic differences often reflect cultural or economic variations, which impact purchasing patterns and sales volume.

By analyzing demographic data, companies can identify high-value customer segments and tailor their sales and marketing strategies accordingly. This segmentation approach helps optimize pricing, promotions, and product distribution to meet the specific needs of each target group.

Conclusion

In today's data-driven marketplace, understanding consumer behavior and trends is essential for driving sales performance. Businesses that successfully interpret customer preferences, respond to influencing factors, adapt to emerging trends, and target the right demographic segments are more likely to achieve sustainable growth.

In your project, this overview provides the conceptual foundation for the analytical part—where real data will be used to explore how these behavioral insights manifest in actual sales performance across different customer groups and regions.



Customer & Demographics KPIs

✦ Number of Unique Customers:

- **Formula:** `COUNT (DISTINCT customer_id)`
- **Insight:** Measures how many distinct customers made purchases during the analysis period.

✦ Average Age of Customers:

- **Formula:** `AVG (age)`
- **Insight:** Identifies the average age of customers to understand the main target demographic.

✦ Sales by Gender (%):

- **Formula:** $(\text{Revenue from Male or Female} \div \text{Total Revenue}) \times 100$
- **Insight:** Shows the percentage of total sales contributed by each gender, highlighting spending differences.

✦ Repeat Customer Rate:

- **Formula:** $(\text{Customers with } >1 \text{ Invoice} \div \text{Total Customers}) \times 100$
- **Insight:** Measures customer loyalty and retention by identifying how many customers made repeat purchases.

✦ Average Revenue per Customer (ARPC):

- **Formula:** $\text{Total Sales} \div \text{Unique Customers}$
- **Insight:** Evaluates the average revenue generated by each customer, useful for understanding overall customer value.



Challenges & Risks in the Sales Industry

Despite technological growth, several **economic and operational challenges** continue to shape sales outcomes. Understanding these challenges helps analysts and decision-makers interpret data trends accurately and design data-driven strategies for improvement.

1. Market Saturation

Definition: Market saturation occurs when most potential customers already own or use similar products, leaving little room for growth.

Impact on Sales:

- Reduced potential for customer acquisition.
- Smaller market shares available for each competitor.
- Pressure on marketing teams to differentiate offerings.

Example Insight:

In a saturated smartphone market, a data analyst might observe declining year-over-year sales despite stable website traffic — a sign that customer conversion is limited by market maturity, not marketing performance.

2. Price Competition

Definition: Price competition happens when multiple businesses sell similar products and compete by lowering prices.



Impact on Sales:

- Thinner profit margins.
- Difficulty maintaining premium pricing strategies.
- Higher dependency on volume sales to stay profitable.

Analytical Approach:

A data analyst may compare **Gross Profit Margin (GPM)** across periods to identify when price cuts have improved sales volume but reduced profitability.

Example:

Sales revenue increased by 10%, but GPM dropped from 45% to 30%, suggesting heavy discounting impacted profitability.

3. Supply Chain Issues

Definition: Supply chain issues refer to disruptions in sourcing, manufacturing, or distribution processes.

Impact on Sales:

- Stock shortages leading to lost sales.
- Increased costs of materials or logistics.
- Delayed product delivery and poor customer experience.

Analytical Insight:

Data analysts often monitor **order completion rates** or **delivery times**. A sudden drop in sales despite high customer demand could indicate stock unavailability due to supply chain delays.

Example:

A 20% decline in completed transactions during Q2 correlates with global shipping delays reported by the logistics department.



4. Changing Consumer Demand

Definition: Consumer preferences evolve with trends, seasons, and external influences (e.g., technology, culture, or economic conditions).

Impact on Sales:

- Outdated products losing popularity.
- Need for continuous product innovation.
- Unpredictable demand affecting inventory and forecasting accuracy.

Analytical Approach:

Tracking **sales by product category** or **customer segment** over time helps analysts identify trends in demand shifts.

Example:

Sales data shows a 40% increase in eco-friendly product purchases, suggesting a consumer shift toward sustainability.

5. Economic or Inflation-Related Challenges

Definition: Economic instability and inflation affect consumer spending power and operational costs.

Impact on Sales:

- Reduced disposable income limits buying behavior.
- Rising costs of production or transportation.
- Profit margins squeezed due to higher expenses.

Analytical Insight:

By analyzing **Average Order Value (AOV)** and **Profit per Transaction**, analysts can measure how inflation influences customer purchase size and profitability.

Example:

Despite stable transaction counts, AOV dropped 12% during inflationary periods, signaling that consumers are purchasing fewer items per order.



Sales Performance Metrics

KPI	Definition	Formula	Insight
Total Sales Revenue	Total income from all sales before costs.	$\Sigma (\text{Price} \times \text{Quantity Sold})$	Measures overall business activity and growth.
Total Profit	Earnings after deducting all expenses.	Revenue – Total Costs	Indicates the company's bottom-line performance.
Gross Profit Margin (GPM)	% of revenue kept after covering product costs.	$(\text{Gross Profit} \div \text{Revenue}) \times 100$	Shows pricing strength and cost control.
Average Order Value (AOV)	Average amount spent per customer order.	Revenue $\div$ Number of Orders	Reveals customer spending behavior.
Profit per Transaction	Average profit made on each sale.	Total Profit $\div$ Number of Transactions	Measures profitability per transaction.



Introduction

This chapter focuses on the **data preprocessing phase**, which includes cleaning the raw dataset and loading it into the **MySQL Workbench** database for further analysis and visualization in Power BI. Data cleaning is an essential step in ensuring data quality, accuracy, and consistency, which directly affects the reliability of analytical results.

Data Overview

Source : Kaggle

Number of rows 99458 and columns

Data base Creation (SQL) + Uploading

Data Base Creation Code

```
CREATE DATABASE sales_performance_db;  
  
use sales_performance_db;  
  
CREATE TABLE Sales (  
    invoice_no VARCHAR(225),  
    invoice_date VARCHAR(225),  
    customer_id VARCHAR(225),  
    gender VARCHAR(225),  
    age INT,  
    category VARCHAR(225),  
    quantity INT,
```




```
selling_price_per_unit DECIMAL(10,2),  
cost_price_per_unit DECIMAL(10,2),  
total_profit DECIMAL(10,2),  
payment_method VARCHAR(225),  
region VARCHAR(225),  
state VARCHAR(225),  
shopping_mall VARCHAR(225),  
PRIMARY KEY (invoice_no, customer_id) );
```

Data Uploading Code

```
SHOW VARIABLES LIKE 'secure_file_priv';  
  
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.4/Uploads/Different  
Store.csv'  
  
INTO TABLE Different_Store  
FIELDS TERMINATED BY ','  
ENCLOSED BY '"'  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS  
  
(invoice_no, invoice_date, customer_id, gender, age, category, quantity,  
selling_price_per_unit, cost_price_per_unit, total_profit, payment_method,  
region, state, shopping_mall);
```



Data Cleaning

Check for Nulls

SELECT

```
SUM(invoice_no IS NULL) AS null_invoice_no,  
SUM(invoice_date IS NULL) AS null_invoice_date,  
SUM(customer_id IS NULL) AS null_customer_id,  
SUM(gender IS NULL) AS null_gender,  
SUM(age IS NULL) AS null_age,  
SUM(category IS NULL) AS null_category,  
SUM(quantity IS NULL) AS null_quantity,  
SUM(selling_price_per_unit IS NULL) AS null_selling_price,  
SUM(cost_price_per_unit IS NULL) AS null_cost_price,  
SUM(payment_method IS NULL) AS null_payment_method,  
SUM(region IS NULL) AS null_region,  
SUM(state IS NULL) AS null_state,  
SUM(shopping_mall IS NULL) AS null_shopping_mall
```

FROM Sales;

Remove duplicates

```
SELECT invoice_no, customer_id, COUNT(*) AS count  
  
FROM Sales  
  
GROUP BY invoice_no, customer_id  
  
HAVING COUNT(*) > 1;
```




Data modeling

```
CREATE TABLE DimCustomer (  
    customer_id VARCHAR(255) NOT NULL, -- Unique, contains letters and numbers  
    gender VARCHAR(10) NULL,          -- Check for NULLs after loading in Power BI  
    age INT NULL,                      -- Check for NULLs after loading in Power BI  
  
    -- Constraints & indexes  
    CONSTRAINT uq_DimCustomer_customerid UNIQUE (customer_id),  
    INDEX ix_DimCustomer_BK (customer_id),  
    INDEX ix_DimCustomer_Lookup (gender, age)  
)  
ENGINE = InnoDB;
```

INSERT INTO DimCustomer (customer_id, gender, age)

```
SELECT DISTINCT  
    customer_id,  
    gender,  
    age  
FROM Sales  
WHERE customer_id IS NOT NULL;
```



CREATE DIMENSION AND FACT TABLES

1- DimDate

```
CREATE TABLE DimDate (  
    date_key INT PRIMARY KEY,      -- Format: YYYYMMDD  
    full_date DATE,  
    year INT,  
    quarter INT,  
    month INT,  
    month_name VARCHAR(20),  
    day INT,  
    day_name VARCHAR(20)  
) ENGINE = InnoDB;
```

2- DimProduct

```
CREATE TABLE DimProduct (  
    product_key INT AUTO_INCREMENT PRIMARY KEY,  
    category VARCHAR(100)  
) ENGINE = InnoDB;
```

3- DimPayment

```
CREATE TABLE DimPayment (  
    payment_key INT AUTO_INCREMENT PRIMARY KEY,  
    payment_method VARCHAR(50)) ENGINE = InnoDB;
```




4- DimLocation

```
CREATE TABLE DimLocation (  
    location_key INT AUTO_INCREMENT PRIMARY KEY,  
    region VARCHAR(100),  
    state VARCHAR(100),  
    shopping_mall VARCHAR(100)  
) ENGINE = InnoDB;
```

5- FactSales (central fact table)

```
CREATE TABLE FactSales (  
    fact_id INT AUTO_INCREMENT PRIMARY KEY,  
    invoice_no VARCHAR(50),  
    invoice_date_key INT,  
    customer_id VARCHAR(255), -- foreign key to DimCustomer  
    product_key INT,  
    payment_key INT,  
    location_key INT,  
    quantity INT,  
    selling_price_per_unit DECIMAL(10,2),  
    cost_price_per_unit DECIMAL(10,2),
```



Foreign key relationships

```
FOREIGN KEY (invoice_date_key) REFERENCES DimDate(date_key),  
FOREIGN KEY (customer_id) REFERENCES DimCustomer(customer_id),  
FOREIGN KEY (product_key) REFERENCES DimProduct(product_key),  
FOREIGN KEY (payment_key) REFERENCES DimPayment(payment_key),  
FOREIGN KEY (location_key) REFERENCES DimLocation(location_key)  
) ENGINE = InnoDB;
```

```
-- POPULATE DIMENSION TABLES FROM Sales
```

-- POPULATE DIMENSION TABLES FROM ECOMMERCE

1- DimProduct

```
INSERT INTO DimProduct (category)  
SELECT DISTINCT  
    category  
FROM Sales  
WHERE category IS NOT NULL;
```

2- DimPayment

```
INSERT INTO DimPayment (payment_method)  
SELECT DISTINCT  
    payment_method  
FROM Sales WHERE payment_method IS NOT NULL;
```




1- DimLocation

```
INSERT INTO DimLocation (region, state, shopping_mall)
```

```
SELECT DISTINCT
```

```
    region,
```

```
    state,
```

```
    shopping_mall
```

```
FROM Sales
```

```
WHERE region IS NOT NULL
```

```
    OR state IS NOT NULL
```

```
    OR shopping_mall IS NOT NULL;
```

2- DimDate

```
-- Generate unique invoice dates
```

```
INSERT INTO DimDate (date_key, full_date, year, quarter, month, month_name,  
day, day_name)
```

```
SELECT DISTINCT
```

```
    DATE_FORMAT(invoice_date, '%Y%m%d') AS date_key,
```

```
    DATE(invoice_date) AS full_date,
```

```
    YEAR(invoice_date) AS year,
```

```
    QUARTER(invoice_date) AS quarter,
```

```
    MONTH(invoice_date) AS month,
```

```
    MONTHNAME(invoice_date) AS month_name,
```

```
    DAY(invoice_date) AS day,
```



```
DAYNAME(invoice_date) AS day_name  
FROM Sales WHERE invoice_date IS NOT NULL;
```

-- POPULATE FACT TABLE

```
INSERT INTO FactSales (  
    invoice_no, invoice_date_key, customer_id, product_key,  
    payment_key,  
    location_key,  
    quantity,  
    selling_price_per_unit,  
    cost_price_per_unit)  
SELECT  
    e.invoice_no,  
    DATE_FORMAT(e.invoice_date, '%Y%m%d') AS invoice_date_key,  
    e.customer_id,  
    p.product_key,  
    pay.payment_key,  
    l.location_key,  
    e.quantity,  
    e.selling_price_per_unit,  
    e.cost_price_per_unit  
FROM Sales
```




LEFT JOIN DimProduct p

ON e.category = p.category

LEFT JOIN DimPayment pay

ON e.payment_method = pay.payment_method

LEFT JOIN DimLocation l

ON e.region = l.region

AND e.state = l.state

AND e.shopping_mall = l.shopping_mall;

SQL Queries for KPIs

-- 1. TOTAL SALES REVENUE -----

-- How much money did the company make from selling all its products?

```
SELECT SUM(selling_price_per_unit * quantity) AS total_sales_revenue  
FROM sales;
```

-- 2. TOTAL PROFIT -----

-- How much money did we keep after product costs?

```
SELECT SUM((selling_price_per_unit - cost_price_per_unit) * quantity) AS  
total_profit  
FROM sales;
```



3. GROSS PROFIT MARGIN

-- Are we pricing our products well?

SELECT

SUM((selling_price_per_unit - cost_price_per_unit) * quantity) AS total_profit,

SUM(selling_price_per_unit * quantity) AS total_revenue,

ROUND(

(SUM((selling_price_per_unit - cost_price_per_unit) * quantity)

/ SUM(selling_price_per_unit * quantity)) * 100, 2

) AS gross_profit_margin_percent

FROM sales;

4. AVERAGE ORDER VALUE (AOV)

-- How much does an average customer spend per purchase?

SELECT AVG(invoice_revenue) AS average_order_value

FROM Sales

-- revenue from each customer invoice

SELECT distinct(invoice_no),

SUM(selling_price_per_unit * quantity) AS invoice_revenue

FROM sales

GROUP BY invoice_no) AS order_revenue;



5. PROFIT PER TRANSACTION

-- How profitable is each sale?

```
SELECT Round(AVG(invoice_profit),2) AS avg_profit_per_transaction
FROM (
    SELECT
        distinct(invoice_no),
        SUM((selling_price_per_unit - cost_price_per_unit) * quantity) AS
invoice_profit
    FROM sales
    GROUP BY invoice_no
) AS total;
```

6. TOP-SELLING CATEGORY

-- Which product type sells the most units?

```
SELECT
    category,
    SUM(quantity) AS total_quantity_sold
FROM sales
GROUP BY category
ORDER BY total_quantity_sold DESC
LIMIT 5;
```



7. CATEGORY PROFITABILITY

-- Which categories are actually making us money?

```
SELECT
    category,
    SUM((selling_price_per_unit - cost_price_per_unit) * quantity) AS total_profit
FROM sales
GROUP BY category
ORDER BY total_profit DESC;
```

8. AVERAGE QUANTITY SOLD PER CATEGORY

-- Which products people buy in bulk vs. single items?

```
SELECT
    category,
    ROUND(SUM(quantity) / COUNT(DISTINCT invoice_no), 2) AS
    avg_quantity_per_transaction
FROM sales
GROUP BY category
ORDER BY avg_quantity_per_transaction DESC;
```




9. CATEGORY REVENUE CONTRIBUTION

-- How much does each category contribute to total sales?

```
SELECT
    category,
    ROUND(SUM(selling_price_per_unit * quantity), 2) AS category_revenue,
    ROUND(
        SUM(selling_price_per_unit * quantity) /
        (SELECT SUM(selling_price_per_unit * quantity) FROM sales) * 100, 2
    ) AS revenue_contribution_percent
FROM sales
GROUP BY category
ORDER BY revenue_contribution_percent DESC;
```

10. AVERAGE MARKUP PER CATEGORY

-- Which products are priced above cost the most?

```
SELECT
    category,
    ROUND(AVG((selling_price_per_unit - cost_price_per_unit) /
        cost_price_per_unit * 100), 2) AS avg_markup_percent
FROM sales
GROUP BY category
ORDER BY avg_markup_percent DESC;
```



11. NUMBER OF UNIQUE CUSTOMERS

-- How many distinct customers have we served?

```
SELECT COUNT(DISTINCT customer_id) AS unique_customers  
FROM sales;
```

12. AVERAGE AGE OF CUSTOMER

-- What's our main age group?

```
SELECT ROUND(AVG(age), 2) AS average_customer_age  
FROM sales;
```

13. SALES BY GENDER

-- Who buys more — men or women?

```
SELECT gender,  
       ROUND(SUM(selling_price_per_unit * quantity), 2) AS total_sales,  
       ROUND(SUM(selling_price_per_unit * quantity) /  
             (SELECT SUM(selling_price_per_unit * quantity) FROM sales) * 100, 2  
             ) AS sales_percentage  
FROM sales  
GROUP BY gender  
ORDER BY total_sales DESC;
```




14. REPEAT CUSTOMER RATE

-- How loyal are our customers?

```
SELECT  
  
    ROUND(  
        (COUNT(DISTINCT CASE WHEN purchase_count > 1 THEN customer_id END)  
        / COUNT(DISTINCT customer_id)) * 100, 2) AS repeat_customer_rate  
  
FROM (  
  
    SELECT  
        customer_id,  
        COUNT(DISTINCT invoice_no) AS purchase_count  
  
    FROM sales  
  
    GROUP BY customer_id  
  
) AS total;
```

15. AVERAGE REVENUE PER CUSTOMER

-- How much revenue does each customer bring?

```
SELECT  
  
    ROUND(SUM(quantity * selling_price_per_unit) / COUNT(DISTINCT  
customer_id), 2) AS average_revenue_per_customer  
  
FROM sales;
```



16. SALES BY REGION

-- Which region sells the most?

SELECT

region,

ROUND(SUM(quantity * selling_price_per_unit), 2) AS total_sales_revenue

FROM sales

GROUP BY region

ORDER BY total_sales_revenue DESC;

17. PROFIT BY SHOPPING MALL

-- Which mall performs best?

SELECT

shopping_mall,

ROUND(SUM((selling_price_per_unit - cost_price_per_unit) * quantity), 2) AS
total_profit

FROM sales

GROUP BY shopping_mall

ORDER BY total_profit DESC;



18. TOP PERFORMING STATE

-- Which state brings the highest revenue?

```
SELECT
    state,
    ROUND(SUM((selling_price_per_unit - cost_price_per_unit) * quantity), 2) AS
    total_profit
FROM sales
GROUP BY state
ORDER BY total_profit DESC
LIMIT 1;
```

19. AVERAGE TRANSACTION VALUE PER REGION

-- How big is an average purchase in each region?

```
SELECT
    region,
    ROUND(SUM(quantity * selling_price_per_unit) / COUNT(DISTINCT invoice_no),
    2) AS avg_transaction_value
FROM sales
GROUP BY region
ORDER BY avg_transaction_value DESC;
```



20. REGIONAL PROFIT MARGIN

-- Which region is the most profitable (not just big sales)?

SELECT

region,

ROUND((SUM((selling_price_per_unit - cost_price_per_unit) * quantity) /
SUM(quantity * selling_price_per_unit)) * 100, 2) AS regional_profit_margin

FROM sales

GROUP BY region

ORDER BY regional_profit_margin DESC;

We Also Tried to calculate those KPIs using measure in PowerBi

1- Total Sales Revenue = SUMX(
'finalproject factsales',
'finalproject factsales'[quantity] * 'finalproject
factsales'[selling_price_per_unit])

2- Total Profit =
[Total Sales Revenue] - [Total Cost]



3- Gross Profit Margin =

```
DIVIDE(  
    [Profit],  
    SUMX(  
        'finalproject factsales',  
        'finalproject factsales'[selling_price_per_unit] *  
        'finalproject factsales'[quantity]  
    )  
)
```

4- Average Order Value =

```
DIVIDE(  
    [Total Sales Revenue],  
    DISTINCTCOUNT('finalproject factsales'[invoice_no]),  
    0  
)
```

5- Profit per Transaction =

```
DIVIDE(  
    [Profit],  
    DISTINCTCOUNT('finalproject factsales'[invoice_no])  
)
```

6- Top-Selling Category =

```
SUMX('finalproject dimproduct', [Total Sales Revenue])
```

7- Category Profitability =

```
CALCULATE(  
    [Total Profit],
```



```
ALLEXCEPT('finalproject dimproduct', 'finalproject  
dimproduct'[category]))
```

8- Average Quantity per Category =

```
AVERAGEX(  
VALUES('finalproject dimproduct'[category]),  
CALCULATE(SUM('finalproject factsales'[quantity]))  
)
```

9- Category Revenue Contribution =

```
DIVIDE(  
[Total Sales Revenue],  
CALCULATE([Total Sales Revenue], ALL('finalproject  
dimproduct'))  
)
```

10- Average Markup per Category =

```
AVERAGEX(  
VALUES('finalproject dimproduct'[category]),  
DIVIDE(  
[Total Profit],  
[Total Cost]  
)  
)
```

11- Unique Customers =

```
DISTINCTCOUNT('finalproject factsales'[customer_id])
```




12- Average Revenue per Customer =
`DIVIDE([Total Sales Revenue], [Unique Customers])`

13- Sales by Region =
`CALCULATE(
 [Total Sales Revenue],
 ALLEXCEPT('finalproject dimlocation', 'finalproject
dimlocation'[region])
)`

14- Profit by Shopping Mall =
`CALCULATE(
 [Total Profit],
 ALLEXCEPT('finalproject dimlocation', 'finalproject
dimlocation'[shopping_mall])
)`

15- Top Performing State =
`VAR StateProfitTable =
 SUMMARIZE(
 'finalproject dimlocation',
 'finalproject dimlocation'[state],
 "TotalProfit", [Total Profit]
)
VAR TopState =
 TOPN(1, StateProfitTable, [TotalProfit], DESC)
RETURN
MAXX(TopState, 'finalproject dimlocation'[state])`

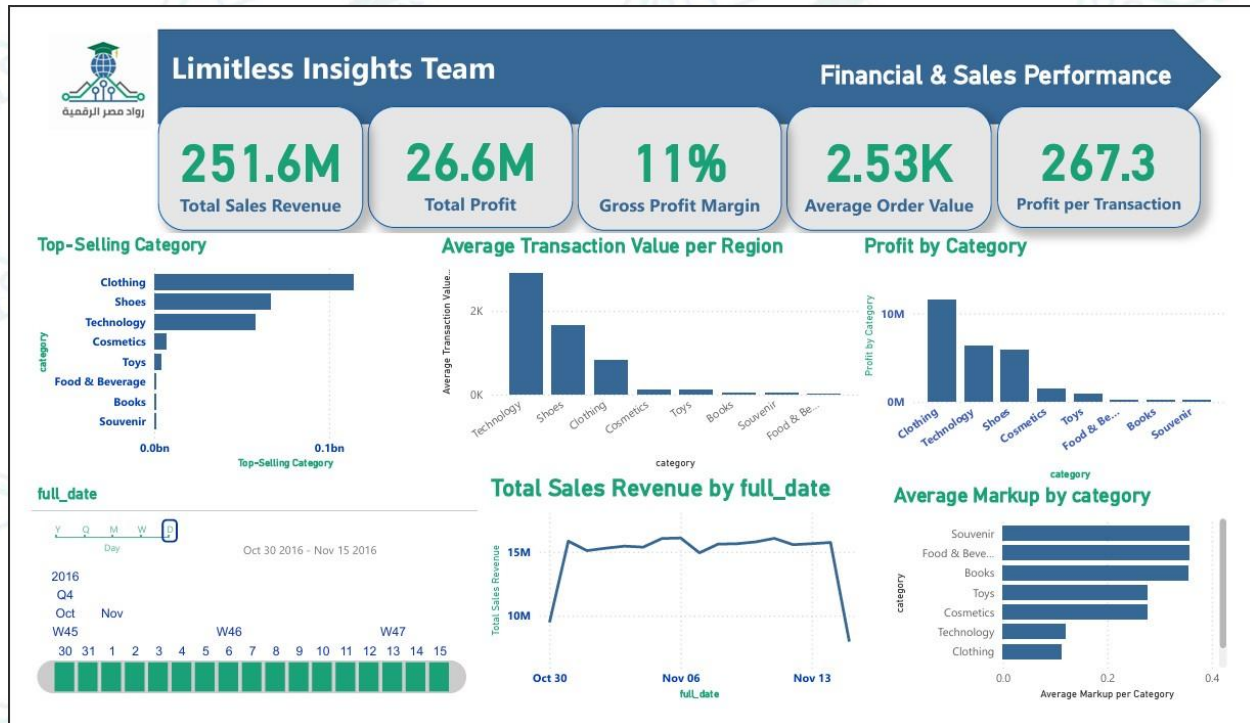


1- Average Transaction Value per Region =
`AVERAGEX(
VALUES('finalproject dimlocation'[region]),
DIVIDE([Total Sales Revenue],
DISTINCTCOUNT('finalproject factsales'[invoice_no])))`

2- Regional Profit Margin =
`DIVIDE(
[Total Profit],
[Total Sales Revenue])`



10. Dashboard & Visualization (PowerBi)





Dashboard Explanation & Insights

Top-Selling Category (Horizontal Bar Chart)

What it shows:

Revenue generated per product category.

Key Insights:

- **Clothing** is the top-selling category by a wide margin.
- **Shoes** and **Technology** follow, performing significantly well.
- Categories like **Books**, **Toys**, **Souvenir**, and **Food & Beverage** are low contributors.

Average Transaction Value per Region (Bar Chart)

What it shows:

Average invoice value per category or region.

Insights:

- **Technology** orders have the highest transaction value (over 2K).
- **Shoes** and **Clothing** follow but at much lower values.
- Items like **Books** and **Food & Beverage** have extremely small average transactions.

Profit by Category (Bar Chart)

What it shows:

Total profit generated by each product category.

Insights:



- **Clothing** is the most profitable category (even more than technology), indicating high markup.
- **Technology, Shoes, and Cosmetics** also perform strongly.
- **Books, Toys, and Souvenir** generate minimal profit.

Date Slicer / Calendar Timeline

What it shows:

A timeline filter allowing the user to select any day, week, month, or year.

Insights:

Enables dynamic filtering of all visualizations to show trends over specific periods.

Total Sales Revenue by Date (Line Chart)

What it shows:

Daily revenue trend over the selected period.

Insights:

- Revenue starts around **10M** and quickly rises to **15M+**.
- Sales remain stable through the period.
- Sharp drop at the end indicates the end of available data (not a real performance drop).

Average Markup by Category (Horizontal Bar Chart)

What it shows:

Average markup percentage for each category.

Key Insights:

- **Souvenir, Food & Beverage, Books, and Toys** have the highest markup (0.2–0.4).
- **Clothing and Technology** have lower markup values.



Average Revenue per Customer by Age (Line Chart)

What it shows:

How average customer revenue changes as customers get older.

Insight:

Revenue tends to vary across age groups with noticeable peaks around mid-30s to mid-40s. These age ranges appear to be the most profitable customers.

Average Quantity per Category (Horizontal Bar Chart)

What it shows:

Average number of items sold per customer in each product category.

Insight:

- **Clothing** is the highest-selling category by quantity.
- Cosmetics and Food & Beverage follow.
- Shoes and Toys contribute the least.

Total Sales Revenue by Gender (Donut Chart)

What it shows:

The percentage and total revenue contribution from male vs female customers.

Insight:

- Male customers contribute ~59.7% of revenue.
- Female customers contribute ~40.3%.

Sales by Region (Horizontal Bar Chart)

What it shows:

Total sales revenue split by region (West, East, Central, South).



Insight:

- **West region** generates the highest revenue.
- East and Central follow closely.
- South performs lower than other regions.

Profit Margin by State (Multi-color Bar Chart)

What it shows:

Profit margin percentage for each state, color-coded by region.

Insight:

- Profit margins are fairly consistent across states.
- No state shows extreme loss or unusually high margins.
- States in the West and Central regions appear slightly stronger overall.

Profit by Shopping Mall (Vertical Bar Chart)

What it shows:

Total profit generated by each shopping mall.

Insight:

- **Kanyon** and **Mall of L...** appear as the top-performing malls.
- Some malls (e.g., Metropol., ViaPort) generate noticeably less profit.

Slicers (Region, State)

What they show:

Filters that let you segment the dashboard by region or specific state.

Insight:

Helps drill down into performance and compare states inside different regions.